

ESSENTIALE - ANIMAL KINGDOM

BASIS OF CLASSIFICATION

1 All members of Kingdom Animalia:

1. have tissues like nerve and muscle.
2. reproduce exclusively by sexual method.
3. are multicellular.
4. exhibit same pattern of organization of cells.

2 Consider the given two statements:

Statement I:	Sponges exhibit a cellular level of organization.
Statement II:	Cnidarians have a tissue level of organization.

In light of the above statements, choose the most appropriate answer from the options given below:

1. **Statement I** is correct but **Statement II** is incorrect.
2. Both **Statement I** and **Statement II** are correct.
3. Both **Statement I** and **Statement II** are incorrect.
4. **Statement I** is incorrect but **Statement II** is correct.

3 A complete digestive system is present in the members of all the following phyla of Animal Kingdom except:

I. Platyhelminthes	II. Mollusca	III. Annelida
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1. Only I	2. Only II
3. Only II and III	4. I, II, and III

4 The type of circulatory system, in which blood is circulated through a series of vessels [arteries, veins and capillaries] and never leaves the network of blood vessels, is called as:

1. An open circulatory system	2. A closed circulatory system
3. A lymphatic system	4. A portal circulation

5 Regarding bilateral symmetry:

I:	Bilateral symmetry is the body plan where the body of the animal can be divided into two identical halves through any plane passing through the central axis of the body.
II:	While Ctenophores and adult Echinoderms have bilateral symmetry, Annelida and Arthropods have radial symmetry.

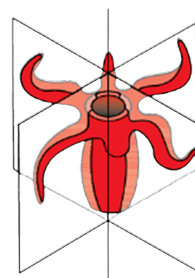
In light of the above statements, choose the most appropriate answer from the options given below:

1. **Statement I** is correct but **Statement II** is incorrect.
2. Both **Statement I** and **Statement II** are correct.
3. Both **Statement I** and **Statement II** are incorrect.
4. **Statement I** is incorrect but **Statement II** is correct.

6 Flatworms will exhibit all the following except:

1. Bilateral symmetry	2. Three embryonic layers
3. Incomplete digestive system	4. Coelom

7 What would be true for an animal that exhibits the kind of body symmetry shown in the given diagram?



I:	This is seen in sponges, cnidarians, ctenophores and larval echinoderms.
II:	This type of body symmetry is generally seen in animals which move actively in their environment.

1. Only I	2. Only II
3. Both I and II	4. Neither I nor II

8 The triploblastic animals differ from the diploblastic animals in having:

1. Ectoderm	2. Mesoderm
3. Endoderm	4. Gastroderm

9 Consider the given features:

A:	Body cavity lined by mesoderm
B:	The mesoderm present as scattered pouches in between the ectoderm and the endoderm

Select the correct option of animals having the features A and B:

	A	B
1.	Aschelminthes	Annelids
2.	Platyhelminthes	Ctenophores
3.	Arthropods	Molluscs
4.	Arthropods	Aschelminthes

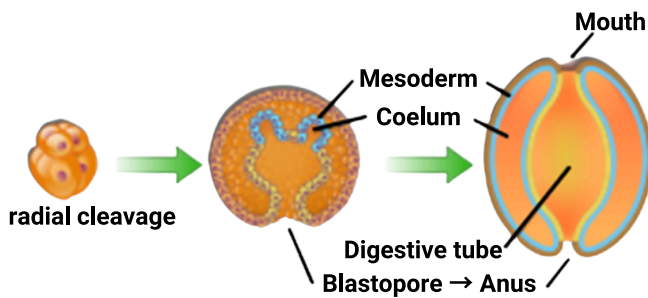
10 In metameric segmentation:

I:	The body is divided into segments only externally.
II:	There is a serial repetition of at least some organs.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

ESSENTIALE - ANIMAL KINGDOM

- 11** Animals with the type of embryonic development shown in the given diagram are:



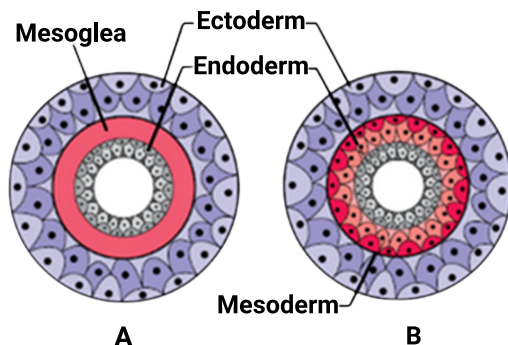
1. Holometabolous
2. Paurometabolous
3. Protostomes
4. Dueterostomes

- 12** The notochord:

I:	is derived from embryonic mesoderm.
II:	is present on the ventral side during embryonic development in chordates.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

- 13** Which of the following will be true for both animals with body plan A and B as shown in the given figure?



1.	Body will be bilaterally symmetrical
2.	Muscle will be well developed
3.	A true coelom will be present
4.	Both will have cells organized into tissues and heterotrophic nutrition

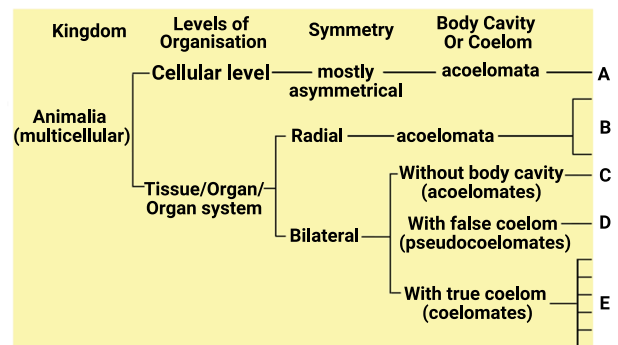
- 14** Consider the two statements:

I:	Metamerism in animals is same as segmentation.
II:	Both result in serial repetition of unit subdivisions of ectoderm and mesoderm products.

1. Both **I** and **II** are correct and **II** explains **I**
2. Both **I** and **II** are correct but **II** does not explain **I**
3. Only **I** is correct
4. Both **I** and **II** are incorrect

CLASSIFICATION OF ANIMALS

- 15** The given figure shows broad classification of Kingdom Animalia based on common fundamental features. Which of the following can be true based on the information given in the figure?



- a:** Sponges represent 'A'
b: 'B' includes flatworms
c: Animals in 'C' are triploblastic
d: Radial symmetry is seen in adults of animals in 'D'
e: Animals in 'E' have a complete digestive system

1.	only a, c and e	2.	only a, b and e
3.	only b and d	4.	only a, c and d

- 16** *Euplectella* [Venus Flower Basket], shown in the figure below, belongs to:



1.	Porifera	2.	Cnidaria
3.	Ctenophora	4.	Platyhelminthes

PHYLUM PORIFERA

- 17** Sponges are mostly:

1.	Asymmetrical	2.	Radially symmetrical
3.	Bilaterally symmetrical	4.	Biradially symmetrical

ESSENTIALE - ANIMAL KINGDOM

18 Which of the following is not a feature of Porifera?

1. Exclusively marine animals
2. Presence of a water transport or a canal system
3. Sexes are not separate
4. Internal fertilization and indirect development

19 Water exits from a sponge through the:

1. Spicule	2. Choanocyte
3. Amoebocyte	4. Osculum

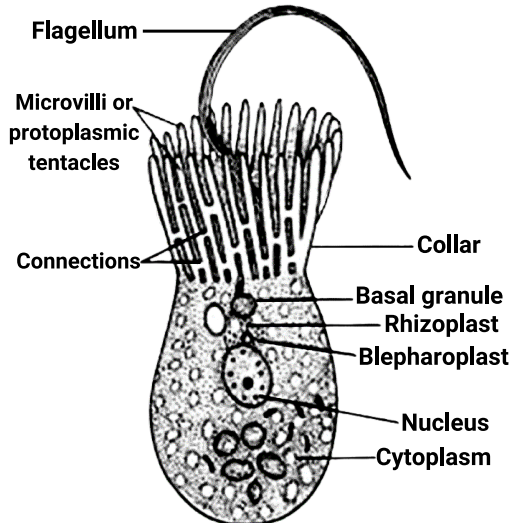
20 Biradial symmetry is most obvious in:

1. Cnidarians	2. Ctenophores
3. Nematodes	4. Echinoderms

21 Body having meshwork of cell, internal cavities lined with food filtering flagellated cells and indirect development are the characteristics of phylum:

1. Porifera	2. Mollusca
3. Protozoa	4. Coelenterate

22 The collared flagellated cell shown in the given figure is called as a:

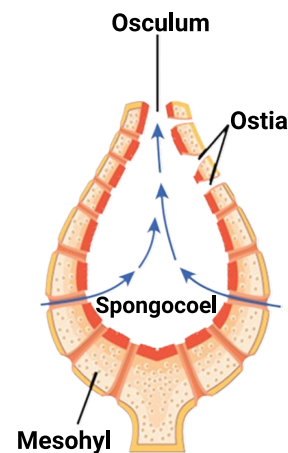


1.	Choanocyte and helps sponges in movement of water through their canal system
2.	Choanocyte and helps sponges in reproducing asexually under favourable conditions
3.	Cnidocyte and helps sponges in catching their prey by paralyzing them
4.	Colloblast and helps sponges in sticking to a surface

23 Identify the incorrect statement regarding sponges:

1.	Water enters the sponge body through ostia into a spongocoel from where it goes out through the osculum.
2.	Body is supported by a skeleton made up of spicules or spongin fibres.
3.	Sexes are not separate.
4.	Fertilization is external.

24 Consider the given two statements regarding the water flow through the bodies of sponges as shown in the diagram.



I:	Most sponges work rather like chimneys: they take in water at the bottom and eject it from the osculum ("little mouth") at the top.
II:	The inner surface is covered with choanocytes where the wave-like motion of the whip-like flagella drives water through the sponge's body.

1. I is correct and II is incorrect.
2. I is incorrect and II is correct.
3. I is correct and II is also correct.
4. I is incorrect and II is also incorrect.

25 Match each item in Column I with one in Column II and select the correct match from the codes given:

	Column I		Column II
A.	<i>Spongilla</i>	P.	Sea-fan
B.	<i>Euspongia</i>	Q.	Brain coral
C.	<i>Gorgonia</i>	R.	Fresh Water sponge
D.	<i>Meandrina</i>	S.	Sea-pen
E.	<i>Pennatula</i>	T.	Bath sponge

Codes:

	A	B	C	D	E
1.	T	R	P	Q	S
2.	R	T	P	Q	S
3.	T	R	Q	S	P
4.	R	T	S	P	Q

ESSENTIALE - ANIMAL KINGDOM

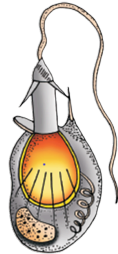
PHYLUM COELENTERATA (CNIDARIA)

26 Regarding the cell shown in the given figure:

I: The name Cnidaria is derived from this cell

II: The stinging capsule present in the cell is called as nematode

III: These cells help cnidarians in anchorage, defence and in capturing the prey



1. While I and II are correct, III is incorrect
2. I, II and III are correct
3. Only II is incorrect
4. Only I is correct

27 Identify the incorrect statement regarding animals belonging to the Phylum Cnidaria:

1. Gastrovascular cavity has a single opening, mouth on hypostome.
2. Corals are cnidarians and have a skeleton of calcium carbonate.
3. They exhibit tissue level of organization and are diploblastic.
4. Digestion is exclusively intracellular.

28 Consider the following statements regarding basic body plans in Cnidaria:

- I:** Polyp is sessile and cylindrical form as seen in *Hydra* and *Adamsia*.
- II:** Medusa is umbrella-shaped and free swimming form as seen in *Aurelia*.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

29 In *Obelia*:

- a:** both polyp and medusa forms exist during its life cycle
- b:** metagenesis, a sort of alternation of generation, is seen
- c:** polyps produce medusae sexually and medusae form the polyps sexually

1. only **a** is correct
2. only **a** and **b** are correct
3. only **c** is correct
4. **a**, **b** and **c** are correct

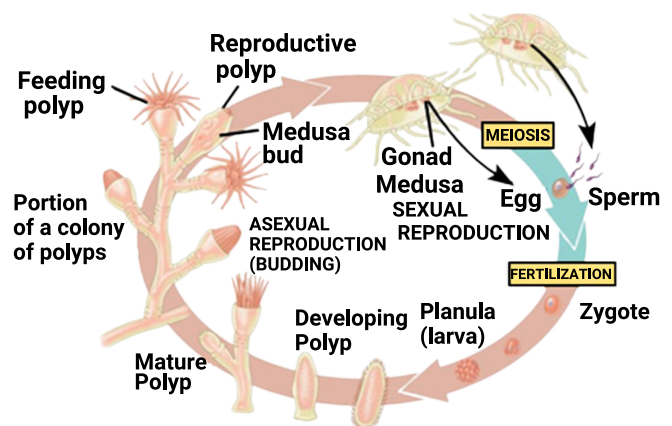
30 Match each item in Column I with their correct match in Column II and select the answer from the codes given:

Column I	Column II
A. <i>Physalia</i>	P. Sea-fan
B. <i>Adamsia</i>	Q. Sea anemone
C. <i>Pennatula</i>	R. Sea-pen
D. <i>Gorgonia</i>	S. Portuguese men of war

Codes:

	A	B	C	D
1.	S	Q	R	P
2.	S	R	Q	P
3.	P	Q	R	S
4.	S	P	Q	R

31 Given below is the schematic representation of the life cycle of *Obelia*, a cindarian. Consider the given two statements:



- I:** This type of life cycle is called as metagenesis and it is a sort of alternation of generation.
- II:** Polyps in the life cycle of cnidarians are haploid and medusae are diploid.

1. Both **I** and **II** are correct and **II** explains **I**
2. Both **I** and **II** are correct but **II** does not explain **I**
3. **I** is correct and **II** is incorrect
4. **I** is incorrect and **II** is correct

PHYLUM CTENOPHORA

32 The phylum with exclusively marine animals is:

1. Porifera	2. Cnidaria
3. Ctenophora	4. Platyhelminthes

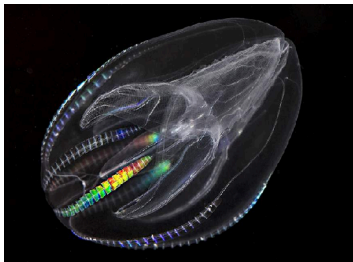
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33 Which of the following is not a feature of Ctenophora?

1. Diploblastic organisms with tissue level of organization
2. Absence of Cnidocytes
3. Presence of internal ciliated comb plates
4. Bio-luminescence well marked in some members

34 The diagram shows an organism with the following features:

- I. Presence of comb plates on the surface.
- II. Bioluminescence



This animal can be:

- | | |
|-----------------|----------------|
| 1. A sponge | 2. A Cnidarian |
| 3. A Ctenophore | 4. A flatworm |

35 In Ctenophora:

- | | |
|-----------|---|
| a: | body bears eight external rows of ciliated comb plates, which help in locomotion. |
| b: | reproduction takes place only by asexual means. |
| c: | digestion is both extracellular and intracellular. |
| d: | sexes are separate. |

The correct statements are:

- | | |
|----------------------------------|----------------------------------|
| 1. a and c only | 2. b and d only |
| 3. a, b and c only | 4. a, b and d only |

PHYLUM PLATYHELMINTHES

36 Flame cells are excretory structures seen in:

- | | |
|--------------------|---------------|
| 1. Platyhelminthes | 2. Annelids |
| 3. Arthropods | 4. Cnidarians |

37 Triploblastic organization, bilateral symmetry but lack of coelom will be seen in:

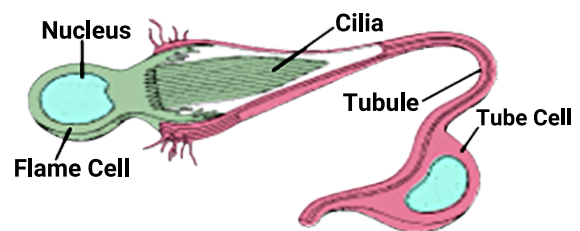
- | | |
|-----------------------|---------------------|
| 1. <i>Planaria</i> | 2. <i>Pheretima</i> |
| 3. <i>Periplaneta</i> | 4. <i>Pinctada</i> |

38 Consider the two statements:

Statement I:	<i>Planaria</i> is a free-living flatworm and possesses high regeneration capacity.
Statement II:	Hooks and suckers are present in parasitic flatworms.

1. **Statement I** is correct and **Statement II** is incorrect
2. **Statement I** is incorrect and **Statement II** is correct
3. Both **Statement I** and **Statement II** are correct
4. Both **Statement I** and **Statement II** are incorrect

39 The structure shown in the given figure helps flatworms in:

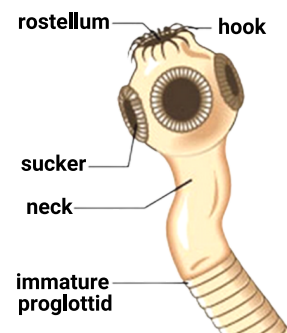


1. Catching food and digesting it
2. Exchange of gases
3. Excretion and Osmoregulation
4. Regeneration of lost body parts

40 Members of the Phylum Platyhelminthes differ from animals belonging to Phylum Annelida and Phylum Arthropoda in:

1. having two germ layers instead of three
2. lacking a coelom
3. showing bioluminescence
4. having a complete digestive system

41 The following structure is most likely:



1. An attachment organ at the anterior end of tapeworms.
2. Sensory apparatus at the anterior end of nematodes.
3. The multiple compound eyes seen in aquatic helminths.
4. Food storage organ in parasitic platyhelminths.

ESSENTIALE - ANIMAL KINGDOM

42 A worm:

I:	has dorso-ventrally flattened body
II:	is an endoparasite in human beings
III:	has hooks and suckers present at its anterior end

In light of the above statements, it will be reasonable to conclude that this worm belongs to Phylum:

1. Platyhelminthes	2. Annelida
3. Aschelminthes	4. Cnidaria

PHYLUM ASCHELMINTHES

43 *Ascaris* is characterized by:

1. Presence of true coelom but absence of metamerism
2. Presence of true coelom and metamerism
3. Absence of true coelom but presence of metamerism
4. Presence of neither true coelom nor metamerism

44 Pinworm, Hookworm, and Filaria worm belong to:

1. Annelida	2. Platyhelminthes
3. Aschelminthes	4. Arthropoda

45 Body cavity is the cavity present between body wall and gut wall. In some animals, the body cavity is not lined by mesoderm. Such animals are called

1. Acoelomate	2. Pseudocoelomate
3. Coelomate	4. Haemocoelomate

46 Consider the two statements:

I:	Platyhelminthes are restricted to environments where dehydration is unlikely.
II:	They are dorso-ventrally flattened.

1. Both **I** and **II** are correct and **II** explains **I**
2. Both **I** and **II** are correct but **II** does not explain **I**
3. Only **I** is correct
4. Both **I** and **II** are incorrect

47 Consider the following statements regarding roundworms:

Statement I:	Alimentary canal is complete with a well-developed muscular pharynx.
Statement II:	An excretory tube removes body wastes from the body cavity through the excretory pore.
Statement III:	They are dioecious and males are longer than females.

The incorrect statement is/are:

1. Only Statement II
2. Only Statement III
3. Only Statement I and Statement II
4. Only Statement II and Statement III

PHYLUM ANNELIDA

48 Metamerism evolved for the first time in the members of the Phylum:

1. Annelida	2. Platyhelminthes
3. Aschelminthes	4. Chordata

49 Unlike Arthropods, Annelids:

1. have a true coelom lined by embryonic mesoderm
2. have a closed circulatory system
3. do not reproduce by sexual methods
4. have a complete digestive system

50 Given below is a figure showing the worms which are the most common cause of helminthic infestation in the world.



1. A is male *Ascaris* and B is female *Ascaris*
2. A is female *Ascaris* and B is male *Ascaris*
3. A is male *Wuchereria* and B is female *Wuchereria*
4. A is female *Wuchereria* and B is male *Wuchereria*

51 Identify the incorrect statement regarding animals belonging to Phylum Annelida:

1.	Aquatic annelids like <i>Nereis</i> possess lateral appendages, parapodia, which help in swimming.
2.	<i>Nereis</i> is dioecious.
3.	Earthworm is monoecious.
4.	<i>Hirudinaria</i> , blood-sucking leech, is an annelid that lacks true coelom and metamerism.

PHYLUM ARTHROPODA

52 Which of the following is not an Arthropod characteristic?

1. Jointed appendages	2. Periodic molting
3. Closed circulation	4. Articulated exoskeleton

ESSENTIALE - ANIMAL KINGDOM

53 Insects are not characterized by:

1. The main body segments – head, thorax and abdomen
2. Three pairs of legs
3. Two pairs of antennae
4. Excretion by Malpighian tubules

54 Identify the correct statements regarding arthropods:

I:	They are bilaterally symmetrical, triploblastic, non-segmented and pseudocoelomate animals.
II:	Arthropoda is the largest phylum of Animalia and includes more than 90 % of all species on the earth that are named so far.
III:	Insects are arthropods with body consisting of head, thorax and abdomen.

1. Only I	2. Only I and II
3. Only III	4. Only II and III

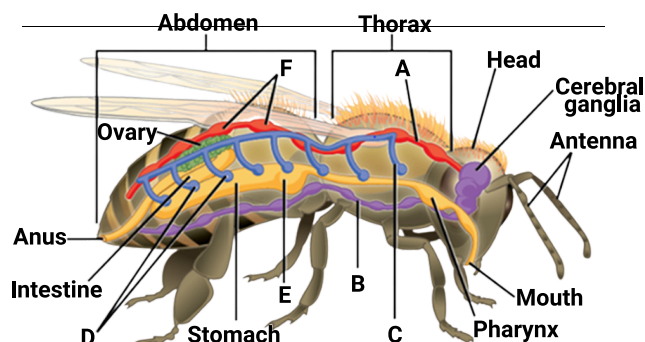
55 Match each item in Column I with one in Column II and select the correct match from the codes given:

Column I	Column II
A. Insect reared for producing silk	P. <i>Locusta</i>
B. Mosquito acting as vector for Dengue virus	Q. <i>Bombyx</i>
C. Gregarious pest	R. <i>Limulus</i>
D. Living fossil	S. <i>Aedes</i>

Codes:

	A	B	C	D
1.	Q	S	P	R
2.	Q	S	R	P
3.	P	Q	R	S
4.	R	P	Q	S

56 In the given diagram of the body plan of an insect, identify the parts A-F:



	A	B	C	D	E	F
1.	Blood vessel	Nerve cord	Trachea	Spiracles	Crop	Hearts
2.	Blood vessel	Nerve cord	Trachea	Spiracles	Oesophagus	Hearts
3.	Nerve cord	Blood vessel	Trachea	Spiracles	Crop	Ostia
4.	Nerve cord	Blood vessel	Stomo-chord	Ostia	Oesophagus	Spiracles

PHYLUM MOLLUSCA

57 Which of the following is unique to the molluscs and has no equivalent in any other animal?

1. Haemocoel	2. Hemocyanin
3. Radula	4. Metamerism

58 Identify the incorrect statement regarding Mollusca:

1.	It is the second largest animal Phylum after Phylum Arthropoda.
2.	It includes animals that are bilaterally symmetrical, triploblastic, and coelomate.
3.	Body is covered with a calcareous shell and is metamerically segmented with a distinct head, muscular foot, and visceral hump.
4.	Gills are present in the mantle cavity, a space between the mantle and the hump.

59 Identify the incorrectly matched pair:

1.	<i>Pinctada</i>	Pearl Oyster
2.	<i>Aplysia</i>	Sea hare
3.	<i>Dentalium</i>	Tusk shell
4.	<i>Loligo</i>	Devil fish

ESSENTIALE - ANIMAL KINGDOM

PHYLUM ECHINODERMATA

60 An important evolutionary transition that appeared for the first time in Echinoderms is:

1. Bilateral symmetry	2. Radial symmetry
3. Tube within a tube plan	4. Endoskeleton

61 A water vascular system is characteristically present in:

1. Annelids	2. Porifers
3. Echinoderms	4. Cartilaginous fishes

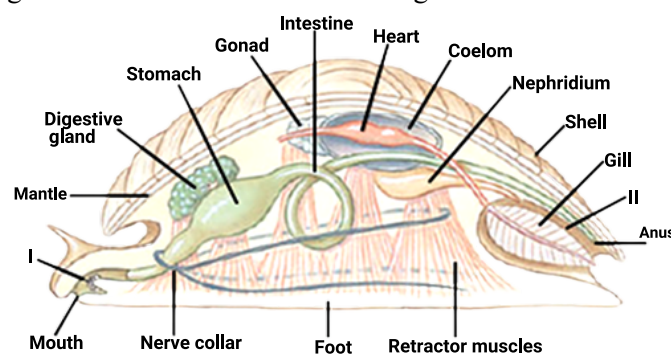
62 Which of the following will be correct regarding Echinoderms?

1. They have bilateral symmetry as adults and radial symmetry as larvae.
2. They are triploblastic and coelomates.
3. They possess a well-developed excretory system.
4. They have an incomplete digestive system.

63 Which of the following is not an Echinoderm?

1. Sea Urchin	2. Sea lily
3. Brittle star	4. Brain coral

64 Given below is the body plan of a hypothetical generalized mollusc. Consider the given two statements.



A:	I is found in all the molluscs.
B:	II is a cavity used for egestion, excretion and respiration.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

PHYLUM HEMICHORDATA

65 In which of the following animals, the body is cylindrical and is composed of an anterior proboscis, a collar and a long trunk?

1. Hemichordates	2. Urochordates
3. Cephalochordates	4. Vertebrate chordates

66 The number of correct statements regarding hemichordates from the given is:

I.	A stomochord is present in the collar region
II.	Circulation is of open type
III.	Excretory organ is proboscis gland
IV.	Development is indirect

1.	1	2.	2
3.	3	4.	4

67 Consider the given two statements:

Statement 1:	Hemichordates were earlier considered as a sub-phylum under Phylum Chordata but are now placed separately in a phylum under non-Chordata.
Statement 2:	Hemichordates have a rudimentary structure in the collar region called stomochord, a structure homologous to the notochord.

1.	Both Statement 1 and Statement 2 are correct but Statement 2 does not explain Statement 1 .
2.	Statement 1 is correct and Statement 2 is incorrect.
3.	Both Statement 1 and Statement 2 are incorrect.
4.	Both Statement 1 and Statement 2 are correct and Statement 2 explains Statement 1 .

68 In hemichordates:

1.	Circulatory system is a closed type
2.	Proboscis gland is a digestive gland that secretes a large number of hydrolytic enzymes
3.	Respiration takes place through the gills
4.	Sexes are not separate, fertilization is internal and development is direct

ESSENTIALE - ANIMAL KINGDOM

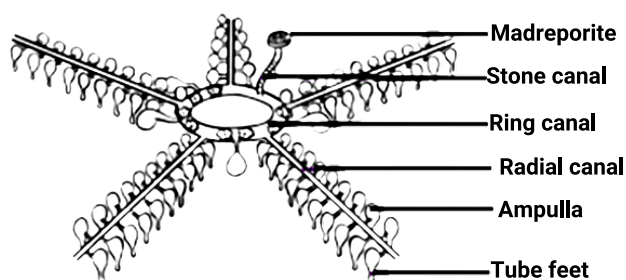
PHYLUM CHORDATA

69 Which of the following features will be seen in chordates but not in non-chordates?

a.	A notochord
b.	Ventral, solid, and double central nervous system
c.	Pharynx perforated by gill slits
d.	Ventral heart
e.	Terminal anus and a lack of post-anal tail

1.	a, c, and d only	2.	a, b, and e only
3.	a and c only	4.	b, c, and d only

70 Given below is a diagram showing the water vascular system of an Echinoderm.



I:	It is a hydraulic system used by echinoderms for locomotion, food and waste transportation, and respiration.
II:	The system is composed of canals connecting numerous tube feet.

1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

71 Consider the given statements:

Statement I:	In Urochordata, notochord is present only in larval tail.
Statement II:	In Cephalochordata, notochord extends from head to tail region and is persistent throughout their life.
Statement III:	Both Urochordates and Cephalochordates are exclusively marine.

Which of the given statements is/ are true?

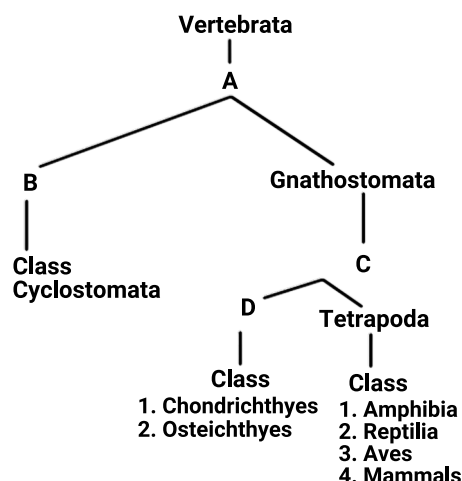
1. Only **Statement I** and **Statement II**
2. Only **Statement I** and **Statement III**
3. Only **Statement II** and **Statement III**
4. **Statement I**, **Statement II**, and **Statement III**

72 Consider the two statements:

Statement I:	All vertebrates are chordates but all chordates are not vertebrates.
Statement II:	Besides the basic chordate features, all vertebrates have at least a three-chambered heart and paired appendages.

1.	Both Statement I and Statement II are correct but Statement II does not explain Statement I .
2.	Statement I is correct and Statement II is incorrect.
3.	Both Statement I and Statement II are incorrect.
4.	Both Statement I and Statement II are correct and Statement II explains Statement I .

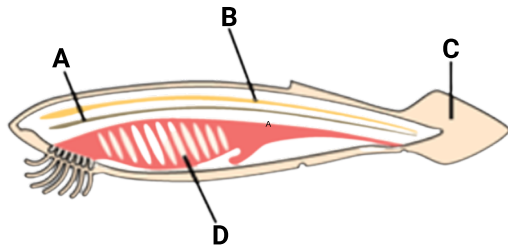
73 The given flow chart shows the classification scheme of sub-phylum Vertebrata. Identify A, B, C and D.



	A	B	C	D
1.	Phylum	Cephalochordata	Urochordata	Protochordata
2.	Division	Agnatha	Super-Class	Pisces
3.	Phylum	Protochordata	Infra-class	Jawless fishes
4.	Division	Agnatha	Sub-division	Pisces

ESSENTIALE - ANIMAL KINGDOM

74 Though all four features shown in the diagram of a generalized chordate appear at some point during development, they are actually named for:



1.	A	2.	B
3.	C	4.	D

CLASS-CYCLOSTOMATA

75 The number of correct statements from the given statements, regarding the members of Class Cyclostomata, is:

- All living members are ectoparasites on some fishes.
- The body bears 6-15 pairs of gill slits for respiration.
- They have a sucking and circular mouth without jaws.
- Their body is devoid of scales and paired fins.
- Cranium and vertebral column are bony.
- Circulation is of open type.

1.	3	2.	4
3.	5	4.	6

76 Regarding reproduction in Cyclostomes:

I:	They are marine but migrate for spawning to fresh water.
II:	After spawning, within a few days, they die.
III:	Their larvae, after metamorphosis, return to the ocean.

- Only **II** is correct
- Only **I** and **II** are correct
- Only **II** and **III** are correct
- I, II** and **III** are correct

77 The organism shown in the given figure is representative of the subphylum Cephalochordata. This organism is:



1. <i>Ascidia</i>	2. <i>Amphioxus</i>
3. <i>Petromyzon</i>	4. <i>Exocoetus</i>

78 Identify the pair of animals that are members of Class Cyclostomata:

1.	<i>Petromyzon, Myxine</i>
2.	<i>Ascidia, Salpa</i>
3.	<i>Pristis, Doliolum</i>
4.	<i>Branchiostoma, Balanoglossus</i>

CLASS – CHONDRICHTHYES

79 Chondrichthyes:

1.	are marine animals and have cartilaginous endoskeleton
2.	have a mouth located dorsally
3.	have teeth that are modified cycloid or ctenoid scales directed backwardly
4.	have the capacity to regulate their body temperature

80 Which of the following is not a cartilaginous fish?

1. Saw fish	2. Dog fish
3. Sea horse	4. Sting ray

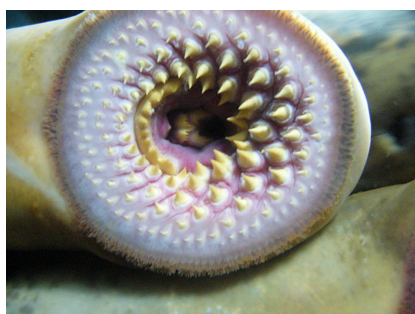
81 What would not be true regarding cartilaginous fishes?

1.	In males, pelvic fins bear claspers
2.	Some of them have electric organs [<i>Torpedo</i>]
3.	None of them is viviparous
4.	Some possess poison sting [<i>Trygon</i>]

ESSENTIALE - ANIMAL KINGDOM

82 The figure shows the mouth of a chordate. It can easily be seen that:

I. The mouth is circular	II. Jaw is absent.
--------------------------	--------------------



This chordate can be:

1. <i>Ascidia</i>	2. <i>Amphioxus</i>
3. <i>Petromyzon</i>	4. <i>Exocoetus</i>

CLASS – OSTEICHTHYES

83 Identify the incorrect statement regarding bony fishes:

I:	They have four pairs of gills which are covered by an operculum on each side.
II:	Due to the absence of air bladder, they have to swim constantly to avoid sinking.
III:	Skin is covered with placoid scales.
IV:	Mouth is mostly terminal.

1. Only II	2. Only II and III
3. Only I and IV	4. Only IV

84 Which of the following is not a bony fish?

1. Angel fish	2. Flying fish
3. Star fish	4. Fighting fish

85 *Exocoetus*, *Labeo*, *Clarias* and *Pterophyllum* are all:

1. endotherms	2. marine fishes
3. protochordates	4. bony fishes

86 Identify the incorrect statement regarding cyclostomes:

1.	Lampreys have direct development, but Hagfishes go through a larval stage followed by metamorphosis into a juvenile form.
2.	Unlike jawed vertebrates, which have three semicircular canals in each inner ear, lampreys have only two and hagfishes just one.
3.	The hagfish blood is isotonic with seawater, while lampreys appear to use the same gill-based mechanisms of osmoregulation as marine teleosts.
4.	Both hagfishes and lampreys have a single gonad.

CLASS – AMPHIBIA

87 In Amphibia:

I:	Skin is moist without scales
II:	A tympanum represents the ear
III:	Fertilization is external

1. Only I and II are correct	2. Only I and III are correct
3. Only II and III are correct	4. I, II, and III are correct

88 *Ichthyophis* is a/an:

1. Tunicate chordate	2. Flightless bird
3. Limbless amphibian	4. Egg-laying mammal

89 Which of the following is not a characteristic of Amphibians?

1.	Most have two pairs of limbs and tail is present in some.
2.	Alimentary canal, urinary and reproductive tracts open into a common chamber called cloaca which opens to the exterior.
3.	Respiration is by gills, lungs and through skin.
4.	The heart is two chambered with one atrium and one ventricle.

90 Which of the following reptilian hearts can be that of a crocodile?

1.		2.	
3.		4.	

CLASS – REPTILIA

91 Which one of the following is not a poisonous snake?

1.	Cobra	2.	Viper
3.	Python	4.	Krait

ESSENTIALE - ANIMAL KINGDOM

92 Most reptiles:

- I. are terrestrial
- II. have a three-chambered heart

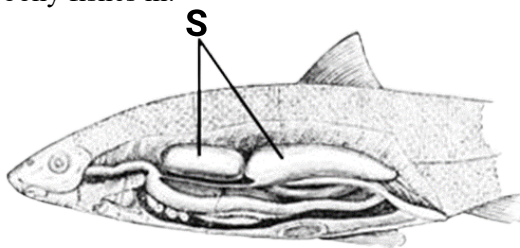
1. Only I is correct	2. Only II is correct
3. Both I and II are correct	4. Both I and II are incorrect

93 In reptiles:

I:	Body is covered by dry and cornified skin, dermal scales or scutes.
II:	External ear is not present and tympanum represents the ear.
III:	Scales are shed as skin cast in all members.
IV:	Fertilization is internal and development is direct.

1. Only II is correct	2. Only II and IV are correct
3. All are correct	4. Only I and II are incorrect

94 The structure marked as 'S' in the given diagram helps bony fishes in:



- 1. Oxygenation of blood
- 2. Achieving neutral buoyancy
- 3. Increasing absorptive surface area for food
- 4. Protection of the nerve cord

95 Which of the following reptiles has a four chambered heart?

1. Crocodile	2. Wall lizard
3. Turtle	4. Tortoise

CLASS – AVES

96 In birds:

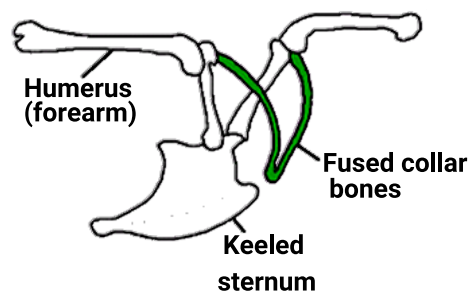
a:	the characteristic feature is the presence of feathers.
b:	forelimbs are modified into wings.
c:	flight is present in all members.
d:	hind limbs do not have scales.

- 1. Only **(a)** and **(d)** are incorrect
- 2. Only **(c)** and **(d)** are correct
- 3. **(a)**, **(b)** and **(c)** are correct
- 4. Only **(a)** and **(b)** are correct

97 Birds are not characterized by:

- 1. a fully ossified endoskeleton
- 2. pneumatic long bones
- 3. absence of crop and gizzard in the digestive tract
- 4. a dry skin without glands except the oil gland

98 The following schematic diagram of a part of skeleton can be that of a/an:



1. Amphibian	2. Reptile
3. Bird	4. Mammal

99 Consider the given two statements:

I:	Birds have a four chambered heart and are warm blooded.
II:	Respiration in birds is by lungs that are connected to air sacs.

- 1. **I** is correct but **II** is incorrect
- 2. Both **I** and **II** are correct.
- 3. Both **I** and **II** are incorrect.
- 4. **I** is incorrect but **II** is correct.

100 Identify the incorrectly matched pair:

	Biological name	Common name
1.	<i>Aptenodytes</i>	Penguin
2.	<i>Pavo</i>	Peacock
3.	<i>Struthio</i>	Pigeon
4.	<i>Psittacula</i>	Parrot

CLASS – MAMMALIA

101 The most unique characteristic of mammals is the presence of:

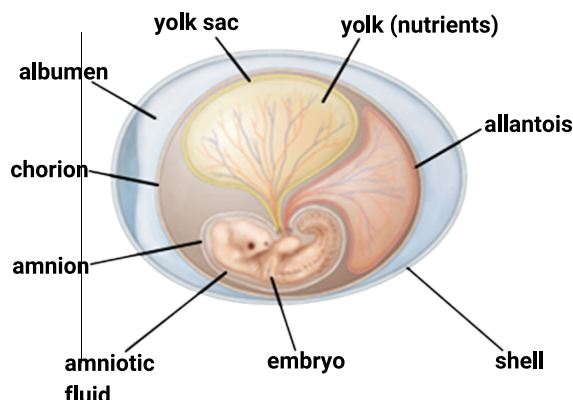
1. a four chambered heart	2. a single aortic arch
3. homeothermy	4. mammary glands

102 Which of the following is an egg laying mammal?

1. <i>Ornithorhynchus</i>	2. <i>Macropus</i>
3. <i>Macaca</i>	4. <i>Balaenoptera</i>

ESSENTIALE - ANIMAL KINGDOM

103 This type of egg cannot belong to a/an:



1.	Amphibian	2.	Reptile
3.	Bird	4.	Mammal

104 Flying fox is a/an:

1.	Flightless bird	2.	Limbless amphibian
3.	Viviparous mammal	4.	Economically important insect

105 Consider the two statements:

I:	The development of an AP axis in animals with bilateral symmetry is important in locomotion.
II:	Bilateral symmetry gives the body an intrinsic direction and allows streamlining to reduce drag.

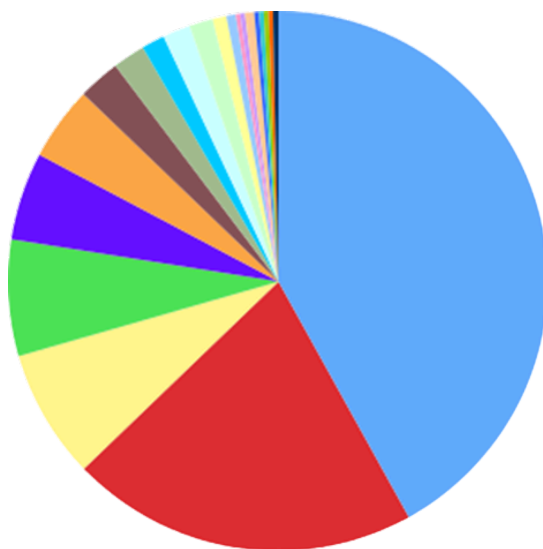
- Both **I** and **II** are correct and **II** explains **I**
- Both **I** and **II** are correct but **II** does not explain **I**
- Only **I** is correct
- Both **I** and **II** are incorrect

106 The bone structure in the following diagram shows:



- Adaptation for flight in birds
- Presence of excess of the hemopoietic tissue in mammals
- Rudimentary limb bones in some reptiles
- Bone of forelegs of frogs

107 As you can see from the given pie chart, over 70% of Mammalia species come from a single order. Which order of class mammalian is this?



1.	Rodentia	2.	Chiroptera
3.	Carnivora	4.	Primata

Answers

1.	(3)	2.	(2)	3.	(1)	4.	(2)	5.	(3)
6.	(4)	7.	(4)	8.	(2)	9.	(4)	10.	(2)
11.	(4)	12.	(1)	13.	(4)	14.	(4)	15.	(1)
16.	(1)	17.	(1)	18.	(1)	19.	(4)	20.	(2)
21.	(1)	22.	(1)	23.	(4)	24.	(3)	25.	(2)
26.	(3)	27.	(4)	28.	(3)	29.	(2)	30.	(1)
31.	(3)	32.	(3)	33.	(3)	34.	(3)	35.	(1)
36.	(1)	37.	(1)	38.	(3)	39.	(3)	40.	(2)
41.	(1)	42.	(1)	43.	(4)	44.	(3)	45.	(2)
46.	(2)	47.	(2)	48.	(1)	49.	(2)	50.	(2)
51.	(4)	52.	(3)	53.	(3)	54.	(3)	55.	(1)
56.	(1)	57.	(3)	58.	(3)	59.	(4)	60.	(4)
61.	(3)	62.	(2)	63.	(4)	64.	(2)	65.	(1)
66.	(4)	67.	(2)	68.	(3)	69.	(1)	70.	(3)
71.	(4)	72.	(2)	73.	(2)	74.	(1)	75.	(2)
76.	(4)	77.	(2)	78.	(1)	79.	(1)	80.	(3)
81.	(3)	82.	(3)	83.	(2)	84.	(3)	85.	(4)
86.	(1)	87.	(4)	88.	(3)	89.	(4)	90.	(3)
91.	(3)	92.	(3)	93.	(2)	94.	(2)	95.	(1)
96.	(4)	97.	(3)	98.	(3)	99.	(2)	100.	(3)
101.	(4)	102.	(1)	103.	(1)	104.	(3)	105.	(1)
106.	(1)	107.	(1)						

ESSENTIALE - ANIMAL KINGDOM

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